OptiCloud AI

Cloud Resource & Cost Optimization Engine

IS CYBERNEXT SUBMISSION

Applicant: Ghadah Alsubaie | **Role:** Cloud & DevOps Engineer | **Target Market:** Saudi Arabia (Nationwide) | **Stage:** Functional MVP

EXECUTIVE SUMMARY

OptiCloud AI is an automated cloud governance and cost-optimization solution developed for enterprise IT, government entities, and cloud engineers. By leveraging cloud APIs, it inspects active workloads, detects idle compute nodes and unattached storage volumes in real-time, and presents dynamic remediation workflows to eliminate financial waste and minimize security attack surfaces.

The Problem

- **Unmanaged Financial Waste:** Cloud environments suffer from 30%–40% operational budget waste due to orphan storage and idle VMs.
- **Expanded Attack Surface:** Forgotten, running compute nodes expose unmonitored endpoints to security threats.
- **Lack of Visibility:** Complex cloud console metrics delay FinOps decision-making.

The Solution & Value

- **API-Driven Automation:** Real-time inspection via GCP Compute Engine APIs to flag underutilized assets.
- **Cost & Risk Reduction:** Instant automated recommendations for instance stopping and disk purging.
- **Vision 2030 Alignment:** Empowers local enterprises to achieve cost-efficient, secure digital transformation.

TECHNICAL ARCHITECTURE & STACK

Python 3.x | Flask Framework | Google Cloud Compute SDK | RESTful APIs | HTML5 / CSS3 | Jinja2 Engine | Git / GitHub | Linux CLI

FUNCTIONAL MVP DASHBOARD PREVIEW

Resource Name	Resource Type	Status	Optimization Action	Est. Monthly Savings
dev-legacy-app-vm	Compute Instance	RUNNING	Schedule Auto-Stop	$65.50
old-backup-snapshot-disk	Unattached Disk	UNATTACHED	Delete / Purge	$34.20
staging-db-replica	Compute Instance	STOPPED	Archive Asset	$42.00

35%+
AVG. COST REDUCTION

Real-Time
API MONITORING

Zero-Trust
ATTACK SURFACE MINIMIZATION